

1. Consider the following LP below wherein the variables x_5 , x_6 and x_7 are the slack variables associated with the three constraints respectively. At the optimal solution, the basic variables are x_7 , x_5 and x_6 where x_7 is the mystery basic variable. **Without using the tableau method of the simplex algorithm**, answer the following questions:
 - a. What is your “guess” of the mystery basic variable at the optimal solution and justify it concisely using concepts learned about the simplex algorithm.
 - b. Determine the values of x_7 , x_5 and x_6 .
 - c. How can you determine if your “guess” is indeed correct?

$$\text{Max } z = 5x_1 + 6x_2 + 4x_3 + 7x_4$$

subject to

$$3x_1 - 2x_2 + x_3 + 3x_4 \leq 135$$

$$2x_1 + 4x_2 - x_3 + 2x_4 \leq 78$$

$$x_1 + 2x_2 + x_3 + 2x_4 \leq 30$$

$$x_j \geq 0, j=1,2,3,4$$

2. Without completely solving any LP show that the maximum value of Z for the following LP cannot exceed 1.

$$\text{Max } Z = x_1 + 2x_2 + x_3$$

subject to

$$x_1 + x_2 - x_3 \leq 2$$

$$x_1 - x_2 + x_3 = 1$$

$$2x_1 + x_2 + x_3 \geq 2$$

$$x_1 \geq 0, x_2 \leq 0, x_3 \text{ unrestricted}$$